

RAJALAKSHMI ENGINEERING COLLEGE

Department of Artificial Intelligence and Data Science

CS23633 – Cloud Computing Laboratory

Experiment No: 2

Title

Deployment of a Flask Web Application using Render (Platform as a Service - PaaS)

Introduction

Cloud computing is a technology that allows users to use computing resources like servers, storage, software, and applications through the internet instead of using only their own computer. It helps developers build and run applications without purchasing or maintaining expensive hardware.

Platform as a Service (PaaS) is one of the cloud service models. In PaaS, the cloud provider gives a complete platform to develop, test, and deploy applications. The provider manages the operating system, server, storage, networking, and software updates, while the developer only needs to write the application code.

In this experiment, a simple web application is created using Flask, which is a lightweight Python web framework. The application contains two pages: a Home page and an About page. The project is stored in GitHub, and Render, which is a PaaS platform, is used to deploy the application. Render automatically installs the required packages, builds the application, and hosts it online, making it accessible through a public URL.

This experiment helps us understand how cloud platforms simplify web application deployment without requiring manual server configuration.

Aim

To develop a simple Flask web application and deploy it on the Render cloud platform using the Platform as a Service (PaaS) model.

Software Requirements

- Windows 10/11
- Python 3.10 or above
- Visual Studio Code
- Git
- GitHub Account
- Render Account
- Internet Connection
- Google Chrome or Microsoft Edge

Input

- Flask application code (**app.py**)
- Dependency file (**requirements.txt**)
- GitHub repository containing the project files

Procedure

Step 1

Install Python 3.10 or later on the system.

Step 2

Install Visual Studio Code and Git.

Step 3

Create a new project folder named **RenderHelloApp**.

Step 4

Inside the project folder, create the following files:

- app.py
- requirements.txt

Step 5

Write the following Flask program in **app.py**.

```
from flask import Flask
```

```
app = Flask(__name__)
```

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```
@app.route("/")
def home():
    return "Hello! Welcome to Render PaaS."

@app.route("/about")
def about():
    return "This application is deployed on Render."

if __name__ == "__main__":
    app.run(host="0.0.0.0", port=5000)
```

Step 6

Add the required packages in **requirements.txt**.

```
Flask==3.0.3
gunicorn==22.0.0
```

Step 7

Open the terminal in Visual Studio Code and install the required packages.

```
pip install -r requirements.txt
```

Step 8

Run the Flask application.

```
python app.py
```

Step 9

Open the browser and test the application locally.

```
http://127.0.0.1:5000
http://127.0.0.1:5000/about
```

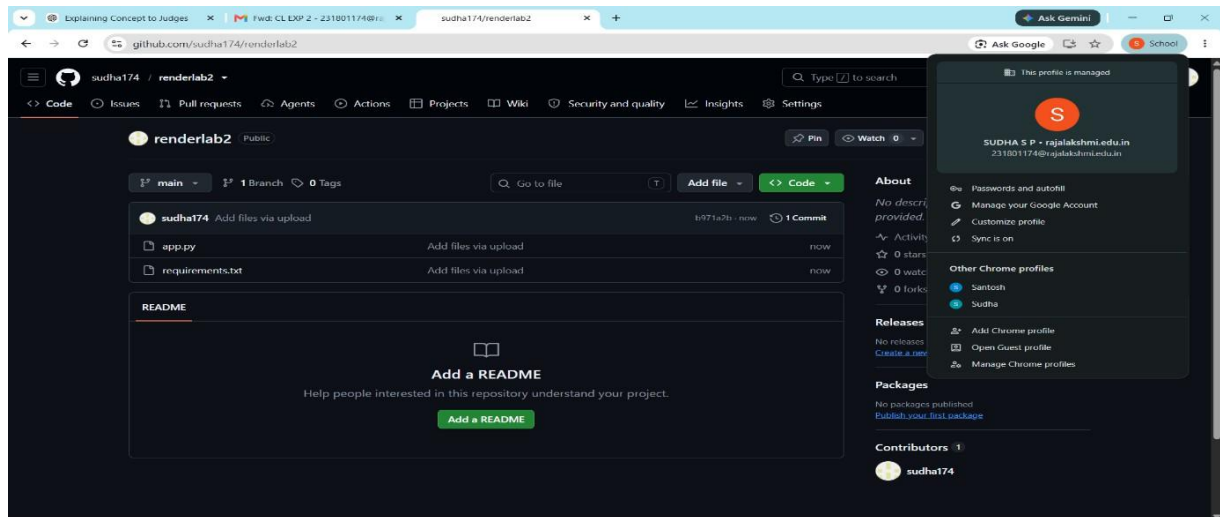
Step 10

Create a GitHub repository named **RenderHelloApp**.

Step 11

Upload **app.py** and **requirements.txt** to the GitHub repository and commit the changes.

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Step 12

Log in to the Render platform using your GitHub account.

Step 13

Create a new **Web Service** and connect it to the GitHub repository.

Step 14

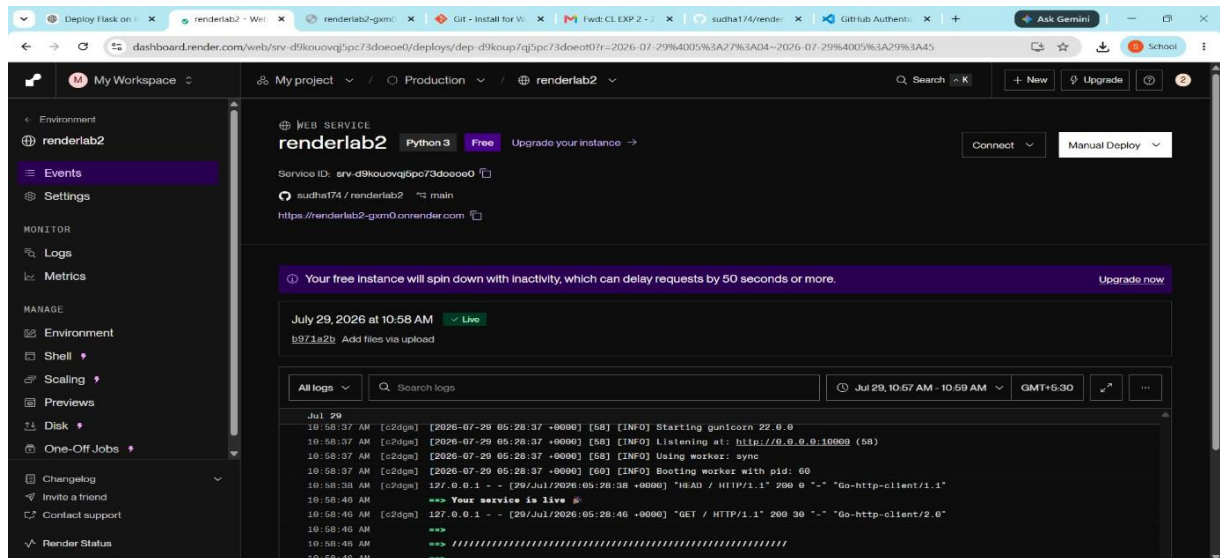
Configure the deployment with the following settings.

Setting	Value
Environment	Python
Branch	main
Build Command	pip install -r requirements.txt
Start Command	gunicorn app:app
Instance Type	Free

Step 15

Click **Deploy Web Service** and wait until the deployment status becomes **Live**.

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Code

app.py

```
from flask import Flask
```

```
app = Flask(__name__)
```

```
@app.route("/")
```

```
def home():
```

```
    return "Hello! Welcome to Render PaaS."
```

```
@app.route("/about")
```

```
def about():
```

```
    return "This application is deployed on Render."
```

```
if __name__ == "__main__":
```

```
    app.run(host="0.0.0.0", port=5000)
```

requirements.txt

```
Flask==3.0.3
```

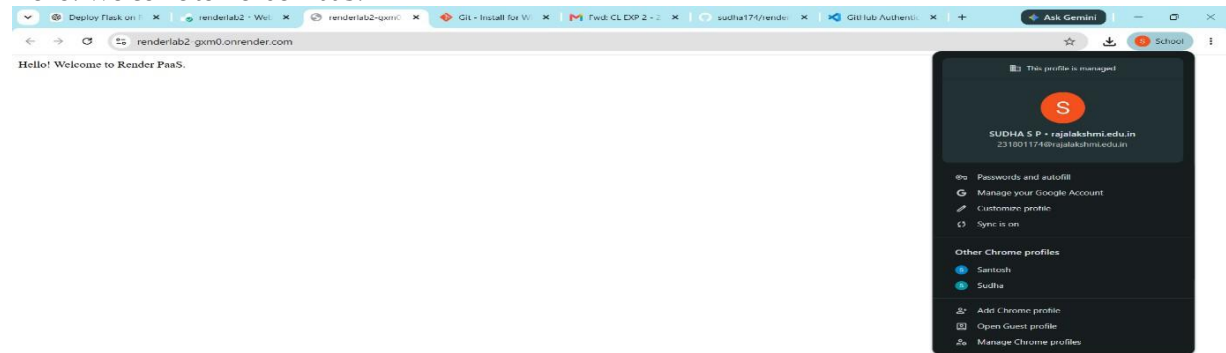
```
gunicorn==22.0.
```

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Output

Home Page

Hello! Welcome to Render PaaS.



About Page

This application is deployed on Render.



Result

The Flask web application was successfully developed and deployed on the Render cloud platform using the Platform as a Service (PaaS) model. The application was tested successfully, and both the Home page and about page were accessible through the public Render URL.

Conclusion

In this experiment, a simple Flask web application was created and deployed successfully using the Render cloud platform. The application was first tested on the local system and then uploaded to GitHub. Render automatically installed the required packages, built the application, and hosted it online. This experiment helped in understanding how Platform as a Service (PaaS) works.